AMD Developer Hackathon • Act II

FusionNet

Your data never leaves. Your model keeps getting smarter.

A federated-learning network where AMD GPUs — from a Low-end PC to an MI300X — collectively fine-tune a shared model without a single byte of private data leaving the device.

ROCm | LoRA Fine-Tuning | Differential Privacy | Zero-Knowledge Proofs

Code & Chaos

THE PROBLEM

Enterprises sit on mountains of sensitive data they cannot legally feed to an LLM

Cloud APIs leak data

Sending patient or financial records to a hosted LLM API violates HIPAA, GDPR, and SOC-2.

On-prem GPUs are unaffordable

A single A100 cluster costs $500K+ — out of reach for most mid-sized enterprises.

Federated learning is CUDA-locked

Flower and FedML are CUDA-first, with no production-ready privacy stack for AMD hardware.

67% of enterprises cite data privacy as the #1 blocker to AI adoption — and no unified solution combines edge inference, privacy guarantees, and AMD-native support.

Why FusionNet Matters

Enterprise AI is increasingly constrained by privacy, compliance, and data residency

Enterprise data is increasingly distributed across edge devices, private clouds, and on-premises infrastructure. Instead of centralizing sensitive data, FusionNet brings AI models to where the data already resides.

The Challenge

- Sensitive data is highly distributed across multiple organizations and silos.
- Strict regulations and legal constraints limit unrestricted data movement.
- Existing solutions require complex integrations of multiple independent tools.

FusionNet's Solution

- **Bring AI to the data, not data to the AI.**
- Privacy-preserving, highly secure federated learning.
- AMD ROCm–accelerated training and edge inference.
- Secure, compliance-safe multi-organization collaboration.

Key Market Drivers

📈 Growing demand for Edge AI

Multi-billion-dollar global market driven by immediate demand for real-time and privacy-preserving AI capabilities.

🔒 Increasing AI regulation

Stricter global standardizations including GDPR, HIPAA, PCI DSS, and the emerging EU AI Act mandates.

🏢 Accelerating enterprise adoption

Modern organizations increasingly require native AI architectures that preserve data privacy while enabling collaboration.

Target Industries & Compliance Standards

🏥 Healthcare
- HIPAA Compliant
- Protects highly sensitive Patient Records

🏦 Finance
- PCI DSS Standard
- Secures vital customer financial transactions

⚖️ Legal
- Attorney Privilege
- Safeguards confidential litigation documents

🏭 Manufacturing
- Edge-Native IIoT
- Processes on-prem factory and operational data

🏛️ Government
- Sovereignty & Trust
- Protects citizen records & critical infrastructure

FusionNet combines privacy-preserving federated learning, secure collaboration, and AMD GPU acceleration into a unified enterprise platform.

THE SOLUTION

Edge devices train locally. Only encrypted, noised deltas ever travel.

AMD Ryzen Laptop

→ Local data → LoRA fine-tune (4-bit)
→ DP-SGD noise added ($\varepsilon \leq 1.0$)
→ Δw encrypted before leaving device

Low-end PC

→ Local data → LoRA fine-tune (4-bit)
→ DP-SGD noise added ($\varepsilon \leq 1.0$)
→ Δw encrypted before leaving device

Radeon GPU

→ Local data → LoRA fine-tune (4-bit)
→ DP-SGD noise added ($\varepsilon \leq 1.0$)
→ Δw encrypted before leaving device

Secure Aggregator — MPC protocol + Zero-Knowledge update verification

AMD Cloud Coordinator (MI300X) FedAvg + privacy noise → updated global model broadcast back to every device

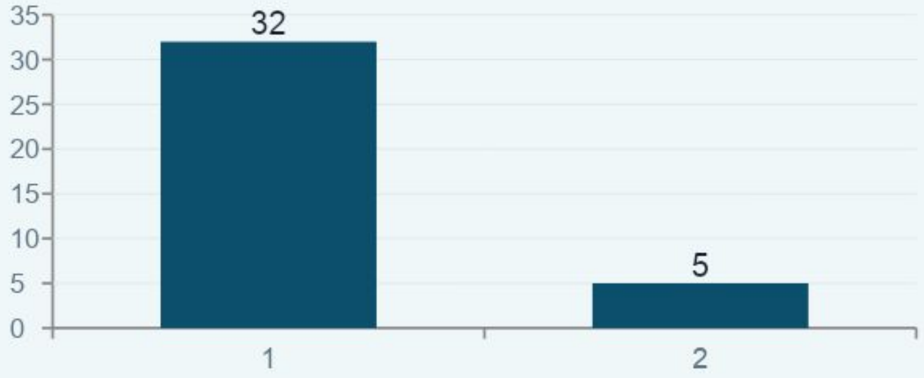
No raw data — patient records, contracts, transactions — ever leaves the originating device.

TECHNICAL DEEP DIVE

Mathematically proven privacy, not just policy promises

AFLoRA on 4-bit Quantized Models

Only a tiny LoRA adapter trains locally; the frozen base model stays 4-bit quantized.



32GB → 5GB of VRAM — feasible on a Steam Deck. B and Λ matrices never leave the device, personalizing locally while A aggregates globally.

Differential Privacy Guarantee

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \times \Pr[M(D') \in S] + \delta$$

- **DP-SGD privacy budget:** $\varepsilon \leq 1.0$ (target)
- **Failure probability:** $\delta \leq$ 1e-5
- **ZKP verification time:** ~2ms per update (Groth16 / PLONK)
- **Secure aggregation:** MPC — SPDZ, no plaintext exposure

Privacy is enforced mathematically — Byzantine-robust aggregation rejects poisoned or out-of-norm updates before they touch the global model.

Built natively on ROCm — not bolted on after the fact

FusionNet FL Orchestrator

PyTorch + ROCm (torch.cuda → torch.hip)

MIGraphX (4-bit inference) + HIP custom kernels (DP noise)

RCCL (collective comms) + rocBLAS

ROCm Runtime

AMD GPU Hardware — MI300X / RDNA3 / Ryzen AI

bitsandbytes

4-bit NF4 quantized inference, native ROCm 6.0+ support

Opacus

Production DP-SGD with a custom fallback for quantized layers

RCCL AllReduce

Cross-device gradient aggregation — drop-in NCCL replacement

Every FusionNet round runs through MI300X cloud nodes and consumer Radeon/Ryzen silicon — the same code path, scaled by device class.

COMPETITIVE LANDSCAPE

The only stack combining AMD-native, full privacy, and heterogeneous edge support

	Privacy Guarantee	AMD Support	Edge Devices	Heterogeneous FL
FusionNet	DP + MPC + ZKP	Native	Yes	Yes
Flower (flwr)	None built-in	Partial	Limited	Partial
FedML	Basic DP	CUDA-first	Limited	Partial
PySyft	Strong	No	No	No
Apple Private FL	Strong	Apple-only	Apple-only	No

FusionNet's unique position: AMD-native + full privacy stack (DP, MPC, ZKP) + heterogeneous edge devices — at once.

Flower and FedML are production-ready but CUDA-first with weak privacy; PySyft and Apple's system have strong privacy but no AMD or heterogeneous support.

BUSINESS MODEL

Five revenue lines, anchored in an AMD ecosystem play

Revenue Stream	Pricing
Enterprise SaaS	$2K–$20K / month
Compute Credits	Rev-share w/ AMD Dev Cloud
Compliance Suite	$5K–$50K / audit report
SDK Licensing (OEM)	$10K–$100K / year
Model Marketplace	15% take rate

TAM
$50B enterprise AI × 30% privacy-blocked = $15B addressable

SAM
Federated learning platforms: $210M ⟶ $1.2B (28% CAGR)

SOM (Yr 3)
100 customers × $60K ACV + AMD OEM deal

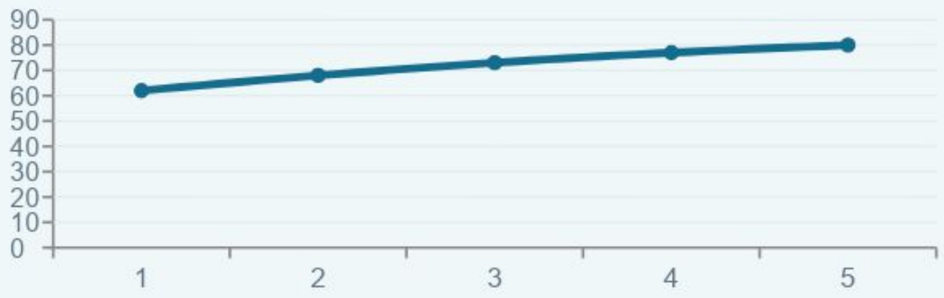
Why AMD would acquire or partner: every FusionNet deployment drives more MI300X cloud usage and Radeon GPU sales — "ROCm Federated" becomes a moat CUDA has no answer for.

Year 3 target: ~$11M ARR ($6M SaaS + $5M AMD licensing)

EXECUTION PLAN

A working MVP now, a four-phase path to $1M ARR

MVP: 10 AMD Cloud VMs · Sentiment Task (2 weeks)



MVP Success Criteria

- ✓ Accuracy +15% from round 0 → 20
- ✓ Zero raw data transmitted (verified via logs)
- ✓ Privacy budget $\varepsilon \leq 1.0$ maintained
- ✓ Graceful 1-VM dropout tolerance

1 Foundation *Mo 1–3*

ROCm-native FL core, DP-SGD accounting, 100 simulated devices

2 Privacy Hardening *Mo 4–6*

Full MPC + ZKP, Byzantine-robust aggregation, first healthcare pilot

3 Edge Deployment *Mo 7–9*

Steam Deck + Ryzen AI clients, 1,000 real edge devices

4 Scale & Commercialize *Mo 10–12*

FusionNet Enterprise SaaS, AMD OEM deal, $1M ARR

Team task tracking, daily checklists, and full risk register live in the project repo.

***“We don’t move data to the model.
We move the model to the data.”***

THANK YOU

First federated-learning system natively optimized for AMD ROCm — your Low-end PC becomes an AI trainer while you sleep.